

Testing a Minimal $N = 16$ Adinkra Completion

Projected one-dimensional linkage equations, covariance tests, and the remaining four-dimensional problem

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Status through 14 July 2026

The project began with a catalog of the 145 positive-dimensional permutation-equivalence classes of length-16 doubly-even codes and a tested implementation of the associated Adinkra calculations. On 14 July, the minimal $128|128$ valise suggested by relaxing the conventional paired auxiliary-spinor count was tested. For catalog entry 75 and dashing class 0, the projected one-dimensional linkage equations have a real solution. A Banach contraction calculation with final rational bounds establishes its existence under the stated arithmetic assumptions. The tested valise does not satisfy any of the higher-dimensional covariance conditions examined here. Its $\text{Spin}(9)$ content is $44 + 84|128$, and therefore does not contain the $9|16$ vector and spinor representations of ten-dimensional super-Yang-Mills theory. After restriction to $\text{Spin}(3) \times \text{Spin}(6)$, the physical gaugino occurs once but the spatial vector and six scalars are absent. The next planned case is a nonvalise four-dimensional gauge and auxiliary complex with maps that remain nonzero at zero momentum.

Scope. No finite off-shell four-dimensional, $N = 4$ or ten-dimensional, $N = 1$ super-Yang-Mills multiplet is claimed. The negative results apply to the tested valise and to the explicitly described local, derivative-only, and direct-sum extensions.

Literature question

The conventional auxiliary-field obstruction traces to Siegel and Roček [1]. The four-dimensional, $N = 4$ problem was later formulated as an Adinkra and linear-algebra completion problem [2, 3]. The paired-unit premise enters the Siegel-Roček count and is stated explicitly in the 2014 reformulation, which asks whether the no-go conclusion changes if that premise is dropped [1, 2]. The present computation provisionally drops only that premise and tests the first resulting $N = 16$ valise size. It does not assume that the unpaired units are admissible Lorentz representations. The calculation also uses the code-to-Adinkra correspondence [4], the published dimensional-enhancement conditions [5], and the recent sixteen-color classification program [6].

Question:

Can a minimal $N = 16$ Garden module contain or encode the physical vector and gaugino so that the resulting representation is finite, local, covariant, and off shell?

The tested valise does not contain the required physical fields as an ordinary covariant subrepresentation or quotient. Gauge and auxiliary complexes with different field content remain untested.

Research baseline through 13 July 2026

Before this investigation, the repository contained the following validated components.

Catalog and representation calculations

The catalog contains 145 permutation-equivalence classes of positive-dimensional length-16 doubly-even codes [7], distributed by code dimension as

k	1	2	3	4	5	6	7	8
classes	4	10	23	38	36	23	9	2

For each dimension- k code class, the catalog records one generator matrix and the calculation constructs its 2^k dashing cohomology classes. Across all 145 code classes, this gives 5,128 pairs consisting of a code class and a dashing cohomology class before height assignments are included. The catalog and representation pipeline constructs the corresponding quotient chromotopologies, dashings, signed-permutation Garden matrices, holoraummy matrices, irreducible decompositions, and Gadget values. The browser interfaces display the catalog and these calculated quantities.

Worksheet and enhancement results

For every catalogued code class, the calculation retains an explicit height assignment and a chirality split with $p, q > 0$. A separate computational verification pass reconstructs each chromotopology from its catalog generators and checks, for all 145 records, the field dimensions, validity of the height assignment, nontriviality of the (p, q) split, and the Gates-Hübsch spin-sum obstruction of Theorems 2.1 and 2.2 [8]. The paper describes evasion of that obstruction as conjectured to be sufficient for worldsheet extension. The present calculation therefore establishes catalog-wide passage of the implemented obstruction. It does not construct the worldsheet transformation laws, prove the sufficiency conjecture, or imply extension to four or ten dimensions. The 145 witnesses are retained in `results/worldsheet_spin_sum_witnesses_n16.json`.

The published dimensional-enhancement equations [5] were first checked against the published $N = 4$ result. At $N = 16$, the calculation used dashing class 0 for each of the 145 code classes and tested a finite candidate set: the valise, one engineered parity-gradient candidate when it defined a valid ranking, and sampled structured source-raised rankings. No tested candidate satisfied the non-gauge ten-dimensional enhancement condition. This conclusion is limited because neither the dashing classes nor the height assignments were searched exhaustively, and the gauge or phantom sector had not been implemented. The cited paper introduces phantoms to handle higher-dimensional gauge invariance.

Pre-cutoff direction

The published Baulieu-Berkovits-Bossard-Martin nine-supercharge construction had been selected as the next calculation involving gauge symmetry [9]. It adds seven auxiliary scalars and closes 9 of the 16 ten-dimensional supercharges off shell. As of 13 July, this calculation had been specified but not implemented.

Minimal-valise analysis on 14 July

Counting premise and exhaustive restricted scans

The reviewed count is

$$16 + 32n = 128m. \quad (1)$$

The increment $32n$ implements the premise that 16-component auxiliary fermionic units occur in pairs. Provisionally dropping only that premise and counting the units individually gives

$$16 + 16q = 128m, \quad (2)$$

whose first solution is $m = 1, q = 7$. Thus $128|128$ is the first size suggested by the relaxed count. This arithmetic does not prove that seven independent auxiliary spinors are Lorentz-covariant, local, or admissible.

For both self-dual length-16 chromotopologies, all 256 dashing classes were tested exhaustively in each of two restricted ansatz classes:

- no literal nine-boson coordinate subspace exists;
- the standard no-leakage embedding has only the zero solution, including arbitrary real mixing.

These two conclusions cover all 512 minimal dashings within the stated ansatz classes. They do not describe an exhaustive solution of the unrestricted bilinear injection and projection equations below.

Projected one-dimensional linkage equations

Introduce linear injection and projection maps

$$\begin{aligned} J_B : \mathbb{R}^9 &\longrightarrow \mathbb{R}^{128}, & P_B : \mathbb{R}^{128} &\longrightarrow \mathbb{R}^9, \\ J_F : \mathbb{R}^{16} &\longrightarrow \mathbb{R}^{128}, & P_F : \mathbb{R}^{128} &\longrightarrow \mathbb{R}^{16}, \end{aligned} \tag{3}$$

with $P_B J_B = I_9$ and $P_F J_F = I_{16}$. These identities make the pairs (J_B, P_B) and (J_F, P_F) linear retractions. The projected one-dimensional linkage matrices are required to equal the nine real symmetric $\text{Spin}(9)$ gamma matrices.

For the $E_8 \times E_8$ topology, catalog entry 75 and dashing 0, a coordinate boson injection and rank-16 fermion projection satisfy one set of linkage equations. For the D_{16} topology, the corresponding support condition has no solution in this ansatz. A coordinate right inverse for the fermion projection does not satisfy the remaining equations. The uniform right inverse also fails: over the rationals, its coefficient matrix has rank 128 and the augmented matrix has rank 129 for each single target component. For the combined nine-target system, the augmented rank is 137.

For catalog entry 75 and dashing class 0, removing those section restrictions leaves 2,304 bilinear equations in 2,863 affine variables. This unrestricted bilinear solve was not repeated for the other 511 minimal dashings. A numerical root has maximum residual 1.11×10^{-15} and a Jacobian of full row rank. A separate a posteriori Banach contraction calculation uses an IEEE-754 round-to-nearest error audit and final rational inequalities. Under those stated arithmetic assumptions, it establishes a real zero within radius 3×10^{-11} of the numerical solution. The final contraction bound is

$$\frac{46133}{3125000000} < 1. \tag{4}$$

The projected one-dimensional system therefore has a real solution. The $112 = 7 \times 16$ dimension of the projection kernel is a dimension count; it does not establish the presence of seven Lorentz-covariant auxiliary spinors.

Covariance and locality tests

Spin(9) equivariance

The sixteen Garden colors carry the physical $\text{Spin}(9)$ spinor action. Its induced action on all 128 bosons and 128 fermions was constructed symbolically. The quadratic Casimirs are

space	Casimir
physical vector	8
physical spinor	9
full bosons	18
full fermions	18

Every equivariant map must intertwine the quadratic Casimir. The unequal eigenvalues force all four maps in the proposed linear retraction to vanish. Thus the real solution of the projected one-dimensional equations is not a $\text{Spin}(9)$ -equivariant linear retraction of the tested field spaces.

Time and spatial derivatives

For a time-derivative map $T(D) = \sum_k T_k D^k$, $\text{Spin}(9)$ commutes with D and the Casimir equation applies coefficient by coefficient. Ordinary node raising and lowering cannot satisfy the required retraction identities over either $\mathbb{R}[D]$ or $\mathbb{R}[D, D^{-1}]$.

For finite local spatial operators over $\mathbb{R}[D, p_1, \dots, p_9]$, evaluation at $p = 0$ gives the already excluded time-derivative map. Hence

$$(P_B J_B)(0, D) = 0 \neq I_9, \quad (P_F J_F)(0, D) = 0 \neq I_{16}. \quad (5)$$

Spatial-potential gauge shifts $\delta A_i = p_i \epsilon$ and Bianchi corrections of positive spatial degree also vanish after setting $p = 0$. The identity therefore fails on the same module.

Spin(9) representation of the valise

Integer characteristic polynomials for four Cartan generators and a separating base-32 combination determine the full joint $\text{Spin}(9)$ character:

$$B_{128} = \text{Sym}_0^2(9) \oplus \Lambda^3(9) = 44 \oplus 84, \quad (6)$$

$$F_{128} = (9 \otimes 16) - 16 = 128, \quad (7)$$

where the fermionic representation is the gamma-traceless vector-spinor. This is the $\text{Spin}(9)$ state content of the eleven-dimensional supergravity multiplet [10]. It does not contain the vector 9 and spinor 16 required by the time-reduced ten-dimensional super-Yang-Mills transformations.

The ordinary ten-dimensional exterior complex was also built symbolically:

$$\Lambda^0 \longrightarrow \Lambda^1 \longrightarrow \Lambda^2 \longrightarrow \Lambda^3, \quad (8)$$

with dimensions 1, 10, 45, 120 and generic ranks 1, 9, 36. Under spatial $\text{Spin}(9)$, the cochains contain

$$1, \quad 1 \oplus 9, \quad 9 \oplus 36, \quad 36 \oplus 84. \quad (9)$$

At total momentum zero, every exterior differential vanishes. A chain-homotopy identity $PJ - I = dH + Hd$ therefore reduces to the ordinary linear retraction already excluded by the Casimir calculation. Replacing potentials by field

strengths does not change this conclusion unless the representation at zero momentum or the differential itself is changed.

An ordinary $N = 16$ valise has boson and fermion dimensions in multiples of 128. The next direct-sum enlargement of the tested representation has dimension 256|256. Direct sums of this representation and its parity reversal contain only combinations of 44, 84, and 128; none contains both a bosonic 9 and a fermionic 16.

Direct four-dimensional covariance test

The validated action was also restricted to $\text{Spin}(3) \times \text{Spin}(6) \simeq SU(2) \times SU(4)_R$. The physical bosons require

$$9 \longrightarrow (3, 1) \oplus (1, 6). \quad (10)$$

Restricted Cartan characters and joint-Casimir nullities give

$$44 \longrightarrow (5, 1) \oplus (3, 6) \oplus (1, 20') \oplus (1, 1), \quad (11)$$

$$84 \longrightarrow (1, 1) \oplus (3, 6) \oplus (3, 15) \oplus (1, 20), \quad (12)$$

and, after complexification,

$$128 \longrightarrow (4, 4) \oplus (4, \bar{4}) \oplus (2, 4) \oplus (2, \bar{4}) \quad (13)$$

$$\oplus (2, 20) \oplus (2, \bar{20}). \quad (14)$$

The real physical gaugino $(2, 4) \oplus (2, \bar{4})$ has multiplicity one. The required bosonic eigenspaces have dimensions

target	found	required
$(3, 1)$ spatial vector	0	3
$(1, 6)$ six scalars	0	6
real gaugino	16	16

Compact-group complete reducibility therefore excludes an ordinary $\text{Spin}(3) \times \text{Spin}(6)$ -equivariant linear retraction from the tested valise onto the physical field representation. The missing representations are bosonic.

Results and scope

One-dimensional result

For catalog entry 75 with dashing 0, the projected one-dimensional linkage equations have a real solution. This establishes consistency of those projected equations only.

Negative results within the tested class

Within the tested valise, the following constructions do not produce the required physical representation:

- literal or standard mixed subspace embedding;
- any $\text{Spin}(9)$ -equivariant linear retraction;
- time node raising or finite local spatial derivatives;
- positive-spatial-degree gauge and Bianchi corrections;
- the ordinary derivative-only exterior gauge complex;
- $256|256$ direct sums of the same valise representation;
- an ordinary $\text{Spin}(3) \times \text{Spin}(6)$ -equivariant linear retraction onto the four-dimensional physical fields.

Scope

The computation does not exclude a different nonvalise or gauge-extended representation, an auxiliary complex with maps that remain nonzero at zero momentum, a rectangular construction, or a nonlocal formulation that removes the zero mode. The last option does not address the usual finite local auxiliary-field problem.

Open questions and next calculations

The next question is whether a finite, local four-dimensional gauge and auxiliary complex at zero momentum can supply the missing off-shell degrees of freedom while respecting $\text{Spin}(3) \times \text{Spin}(6)$, engineering parity, and the physical cohomology. The following calculations address that question in order:

- Algebraic complex at zero momentum: enumerate allowed gauge, compensator, Bianchi, and auxiliary maps at zero momentum, then compute their cohomology over the chosen coefficient field.
- Missing-representation inventory: if the zero-momentum complex fails, calculate the missing $\text{Spin}(3) \times \text{Spin}(6)$ representations and the smallest admissible field content before choosing another module size.
- Nonvalise or rectangular one-dimensional linkage: build the smallest candidate implied by that inventory and require one-dimensional closure without equations of motion.
- Full spatial closure: add all three spatial derivative linkages and classify every remainder as a translation, gauge transformation, a term in the image of the complex differential, or an identity.
- Lorentz and internal-symmetry covariance: construct rotations, boosts, and the required internal symmetry on fields, supercharges, gauge parameters, auxiliaries, and differentials.
- Finite, local, off-shell verification: exclude inverse derivatives, hidden boundary conditions, equations of motion, and infinite auxiliary towers.
- Invariant abelian action: construct a local quadratic action, prove invariance under all sixteen supercharges, and verify the physical spectrum and auxiliary signs.
- Counting-premise assessment: after closure and action invariance, compare the construction with the published counting premises and determine which assumption, scope condition, or inference requires revision. Confirm that the construction preserves locality, finiteness, covariance, and off-shell closure.
- Nonabelian deformation: promote the validated abelian system order by order in the coupling and repeat the closure and action tests.

Inverse spatial derivatives, deletion of zero-momentum modes, boundary-condition-dependent constructions, and direct ten-dimensional extension are separate cases. None resolves the stated finite, local, four-dimensional problem.

The detailed decision tree and acceptance and rejection criteria are recorded in the project roadmap included with the repository.

Computational artifacts and reproduction

The tested command-line pipeline performs the catalog construction, representation calculations, minimal-valise search, and contraction-bound verification. Standalone programs perform the symbolic and numerical audits listed below. The retained outputs and corresponding reproduction commands are given below.

Repository: github.com/p1p3dream/adinkra-codespace.

Artifact	Recorded result
<code>results/sr_hole_minimal.json</code>	Arithmetic, topology, dashing, and minimal embedding calculation.
<code>results/worldsheet_spin_sum_witnesses_n16.json</code>	Explicit height and chirality records for all 145 code classes. The <code>worldsheet-verify</code> command performs the catalog, dimension, height, chirality, and spin-sum checks.
<code>results/sr_joint_root_audit.json</code>	Banach contraction bounds for the projected linkage equations.
<code>results/sr_so9_equivariance_audit.json</code>	Symbolic field action, covariance, Lie closure, and Casimir obstruction.
<code>results/sr_nonvalise_locality_audit.json</code>	Time-derivative ring obstruction.
<code>results/sr_spatial_gauge_locality_audit.json</code>	Zero-spatial-momentum locality result for the stated class of gauge and Bianchi corrections.
<code>results/sr_spin9_decomposition_audit.json</code>	Verified $44 + 84 128$ character.
<code>results/sr_gauge_complex_audit.json</code>	Exterior complex, chain-homotopy condition, and $256 256$ direct-sum calculation.
<code>results/sr_spin3_spin6_branching_audit.json</code>	Verified direct four-dimensional branching and physical-field multiplicities.

```

cargo run --release -- sr-investigation
cargo run --release -- worldsheet-verify
cargo test
python3 scripts/verify_sr_uniform_section.py
python3 scripts/search_sr_joint_section.py --starts 4 --iterations 100
python3 scripts/prove_sr_joint_root.py
python3 scripts/check_sr_so9_equivariance.py
python3 scripts/check_sr_nonvalise_locality.py
python3 scripts/check_sr_spatial_gauge_locality.py
python3 scripts/check_sr_spin9_decomposition.py
python3 scripts/check_sr_gauge_complex.py
python3 scripts/check_sr_spin3_spin6_branching.py

```

At the date of this note, the project test suite reports 239 passed tests, no failures, and six ignored slow tests.

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